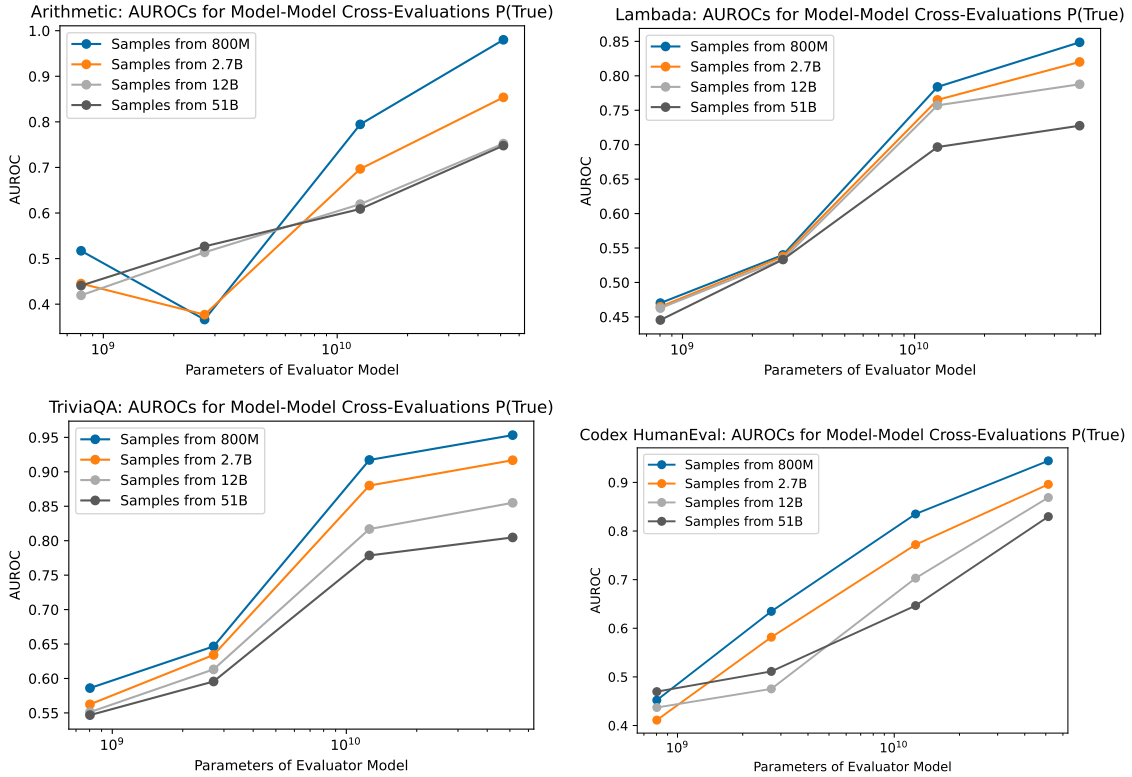

Figure 33 We show AUROCs for P(True) from cross-evaluation experiments where we generate samples from models of one size and then evaluate whether these samples are correct using models of a different size. We see that samples from smaller models are always easier to evaluate, and that larger models are better at evaluation. That said, correctness evaluation improves with the size of the evaluator model, even when it is evaluating its own samples.

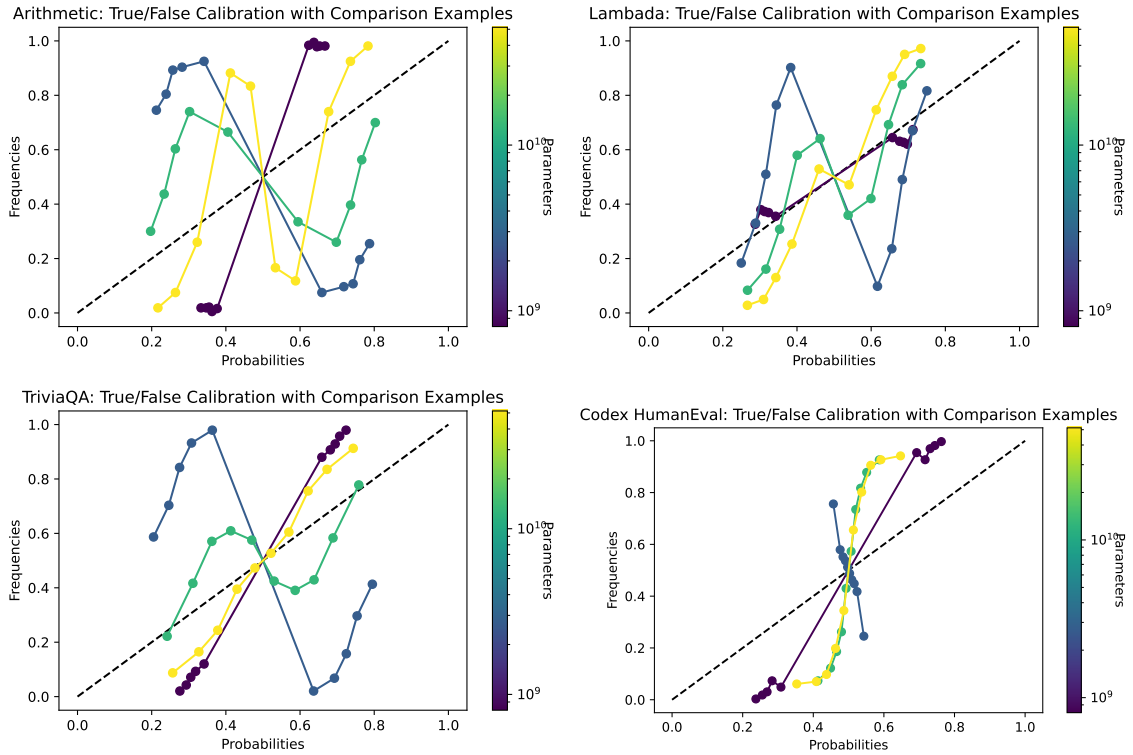

Figure 34 We show that when we ask models to self-evaluate their own samples as True or False **zero-shot**, the P(True) judgment is poorly calibrated, though in most cases it seems to be improving as models become larger and more capable.

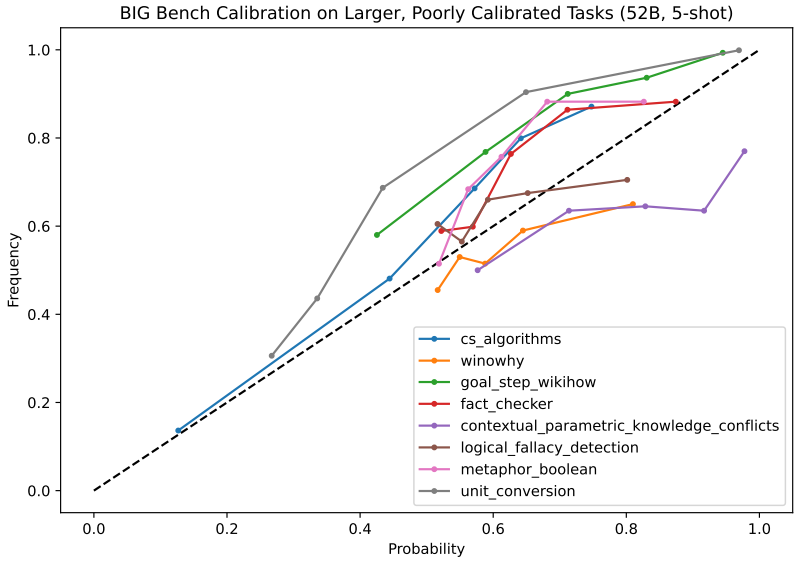

Figure 35 We show calibration curves for a subset of tasks that have both a large number of questions (for better statistics) and fairly bad calibration.

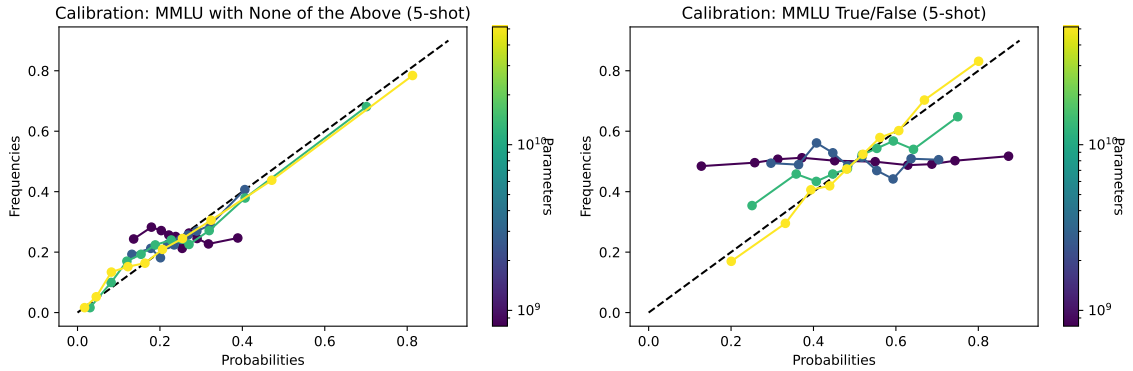

Figure 36 **(left)** We show calibration curves for MMLU when we replace option (D) with 'none of the above' in all questions, or **(right)** when we completely reformulate the task as a True/False decision as to whether randomly selected options are correct.

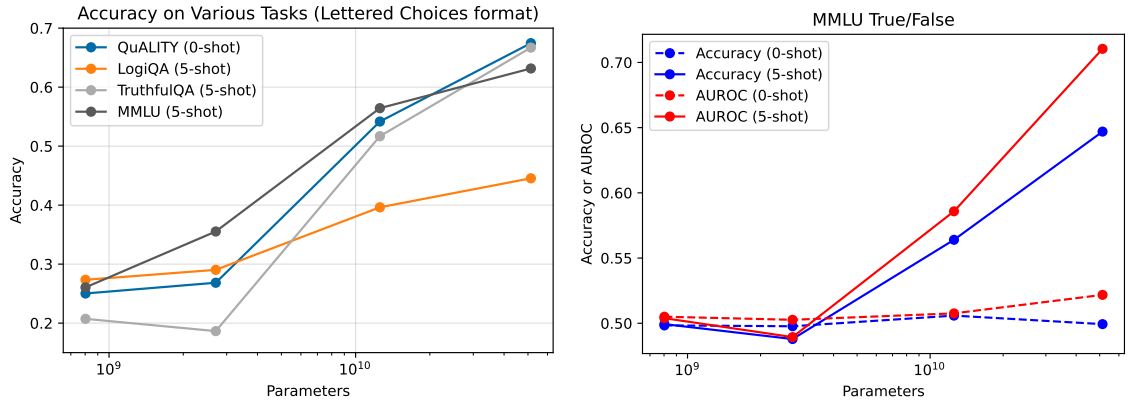

Figure 37 **(left)** We show accuracies on the specific multiple choice tasks we evaluate, MMLU, TruthfulQA, QuALITY, and LogiQA. **(right)** We show accuracy for MMLU in the True/False format, where the model must identify if a possible response to a question is true or false. Chance accuracy is 0.5. In order to connect with other methods that do not produce calibrated probabilities, we also show the AUROC when using the probabilities assigned to True/False to separate the correct and incorrect responses.